

Fleet Maintenance Optimization:

A Comprehensive Cost & Event Analysis (Jan 2023 – June 2024)

Introduction

When I first opened this dataset of over 25,000 vehicle service logs, my mind went blank. I honestly thought, *“What am I even going to find here?”* The data seemed too limited at first glance. There weren't many models, and without vehicle registration numbers, I couldn't even pinpoint exactly which individual vehicle was burning the most cash.

But then it clicked.

I noticed the date segment, meaning I could track exactly which months spiked in costs. Then I saw the other variables: road types, climate conditions, maintenance types, and service levels. Suddenly, my vision expanded. I realized I was looking at a massive combination of facts that could tell a hundred different stories.

So, welcome to my fleet project. This isn't a flashy, entertaining data story. It's just the raw reality of a 260M fleet, broken down into its true 6-month chapters. Let's dive in.

Project Toolkit

To process over 25,000 service logs and uncover these patterns, I built an end-to-end data pipeline using this modern analytics stack:

- BigQuery & SQL: Heavy-duty data warehousing and root-level view optimization to query millions in fleet costs instantly.
- Excel & Power Query: Initial data cleaning, transformation, and structural shaping before injection into the warehouse.
- Power BI & DAX: Advanced time-intelligence modeling, interactive cross-filtering, and executive-level KPI dashboards.
- GitHub: Version control and public repository hosting for project deployment transparency.

Project Assets

- Live Repository: [GitHub Project Space](#)
- RawDataset: [Fleet_Managment_Project/FLEET_MANAGMENT_PROJECT/MAIN_DATA_SOURCE at main · abhishek862000/Fleet_Managment_Project · GitHub](#)

Page 1: Vehicle details and maintenance reports

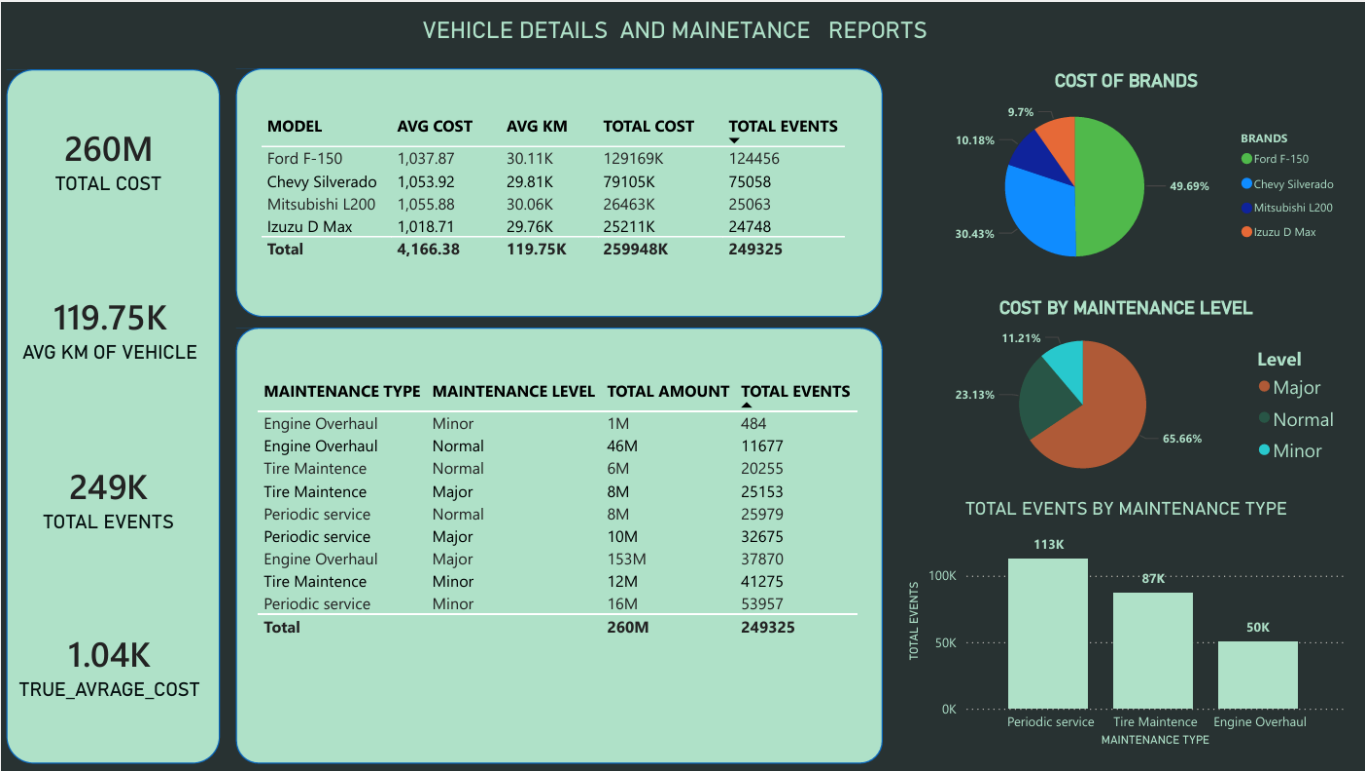


Fig 1.1

The 260M Reality Check

Looking at the main dashboard view in , the high-level totals are massive:

- Total Cost: 260M
- Total Events: 249K maintenance check-ins
- True Average Cost: 1.04K per single event

Here is the breakdown by the numbers so it is easy to catch.

1. The Vehicle Models (Who is spending?)

- Ford F-150 (The King of Expenses):
 - 129M total cost (almost half the entire budget)
 - 124K total events
- Chevy Silverado:
 - 79M total cost
 - 75K total events
- Mitsubishi L200 (Highest Avg Cost):
 - 26M total cost
 - 25K total events
 - 1,055.88 highest average cost per service
- Isuzu D-Max:
 - 25M total cost
 - 25K total events

The Data Blindspot: The tables in Fig 1.1 group everything cleanly by brand, but they hide specific vehicle IDs or registration numbers. We can see *which model* burns money, but we

cannot pinpoint the exact rogue trucks destroying the budget. Future data sampling is required to fix this.

2. Maintenance Types (What are we paying for?)

- Engine Overhauls (The Cash Killer):
 - 200M total cost (swallows the vast majority of the budget)
 - 50K events (*lowest event count, but highest cost*)
- Periodic Services:
 - 34M total cost
 - 113K events (*highest event count, but cheap*)
- Tyre Maintenance:
 - 26M total cost
 - 87K events

3. Maintenance Intensity (How severe is it?)

- Major Maintenance: 171M total cost | 96K events
- Normal Maintenance: 60M total cost | 58K events
- Minor Maintenance: 29M total cost | 96K events

The Bottom Line

Almost half our money goes directly to Ford F-150s, and 200M of our total 260M cash is completely swallowed up by engine overhauls. We know *what* is costing us money; our next step is using the monthly interactive features to track *when* and *where* to stop the bleeding.

Page 2 : 2023 Jan - 2024 June cost analysis by month

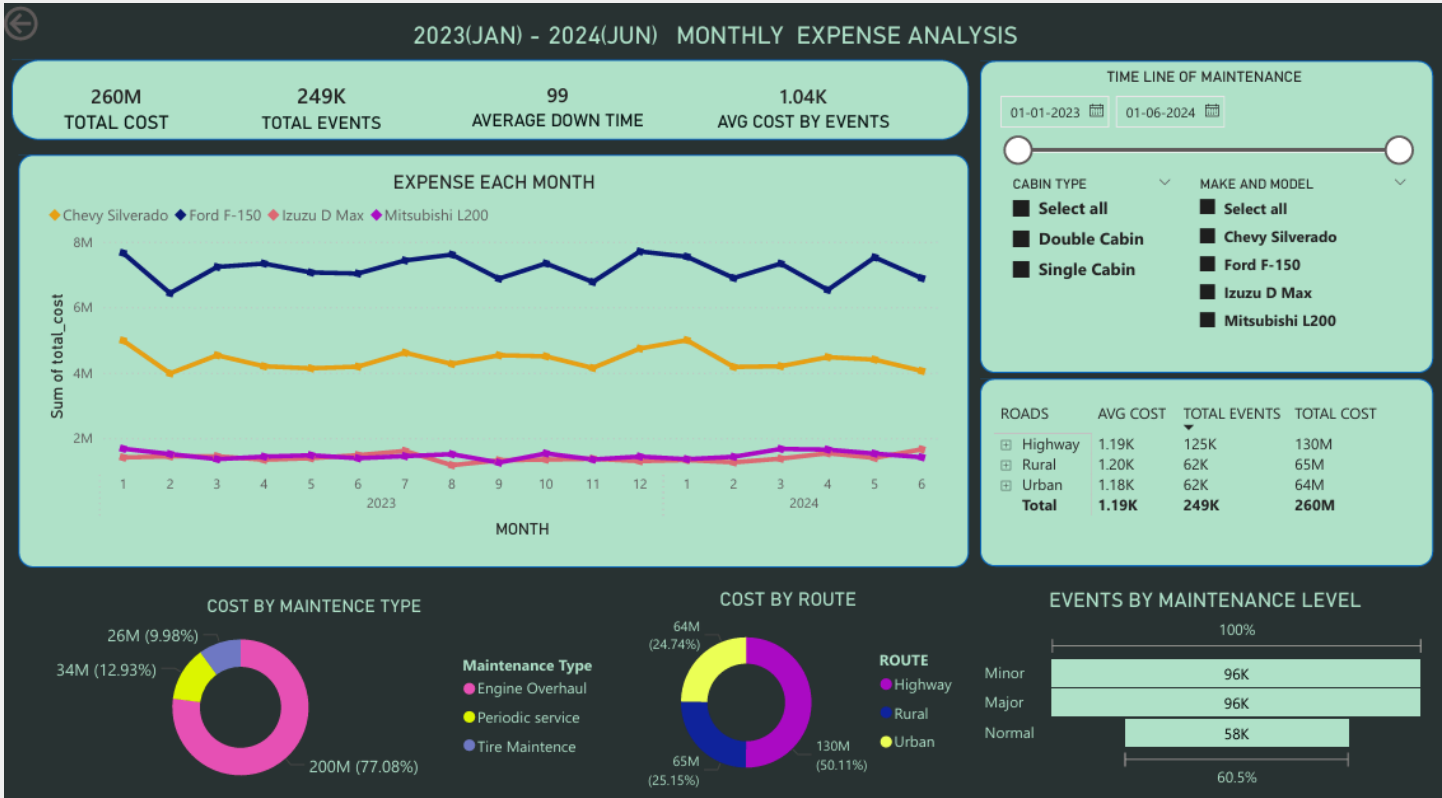


Fig 2.1

Page 2: The Monthly Breakdown

This page is built for interactive filtering. As shown in image_c1e481.png, you can use the timeline slider, select cabin types, or filter by routes to instantly update the KPIs and charts. Here are the hard facts, broken down by the numbers.

1. Double Cabin vs. Single Cabin (Body Type Performance)

- Double Cabins:
 - 144M total cost
 - 138K events
 - 100 hours average downtime
 - 1.04K average cost per service
- Single Cabins:
 - 116M total cost
 - 112K events
 - 99 hours average downtime
 - 1.04K average cost per service

2. The Monthly Burn (Steady Brand Spending)

The monthly line graph trends remain identical every single month:

- Ford F-150: 7M – 8M per month (*Always highest*)
- Chevy Silverado: 4M – 5M per month (*Steady second*)
- Isuzu D-Max & Mitsubishi L200: 1M – 2M per month (*Flat at the bottom*)

3. The 6-Month Copy-Paste (Timeline Intervels)

The fleet behaves almost identically across every 6-month chapter:

- Jan–Jun 2023: 86M total cost | 82K events | 43M highway cost

- Jul–Dec 2023: 87M total cost | 83K events | 43M highway cost
- Jan–Jun 2024: 87M total cost | 84K events | 43M highway cost
- (*Baseline shown in i fig 2.1*)

4. Quick Operational Facts

- | | |
|--------------------------|---------------------|
| • Route Costs: | • Climate Costs: |
| Highway: 130M total cost | Clear: 69M (79.09%) |
| Rural: 65M total cost | Rainy: 14M (15.62%) |
| Urban: 64M total cost | Snowy: 8 M (3.1%) |
| | Windy: 5 M (2%) |

Conclusion: The Story the Numbers Told

When I started this, I thought I was just looking at a massive pile of 25,000 logs. I was worried the lack of registration numbers would keep the real story hidden. But after digging through the 260M spend, the pattern is actually very clear: this fleet is remarkably predictable.

It's a machine that demands roughly 87M every six months like clockwork, regardless of the season or the year. We now know exactly where the pressure points are—the Ford F-150s are the workhorses, the highways are the primary battleground, and the Major Engine Overhauls are the financial anchors that swallow 200M of our budget.